

Abacus row packing

Abacus-lite processes cells by increasing x and tries every row

The idea

- Cross-row spread beats single-row shelf pack on displacement
- Abacus is the detailed legalizer in this course: more search, less movement

Pseudocode

- Abacus needs a nested loop in pseudocode: outer cells in x order, inner trial of every row
- For each trial you compute leftmost legal x and an L1 cost back to the origin
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Fixed macros sit first so their intervals block later trials
- On the overlap seed the sketch lands A at four two

Algorithm sketch — try these

```
INPUT: origin, widths, rows Y[], fixed macros
OUTPUT: legal pack minimizing  $\Sigma$  L1 move
place fixed macros first
order  $\leftarrow$  movables by origin.x
for each cell c in order:
    for each row y: trial leftmost legal x
    keep (x,y) with min  $|\Delta x| + |\Delta y|$  to origin
    place c at best
GOLDEN: A(4,2) B(4,0) C(4,4); disp=4; HPWL=38
```

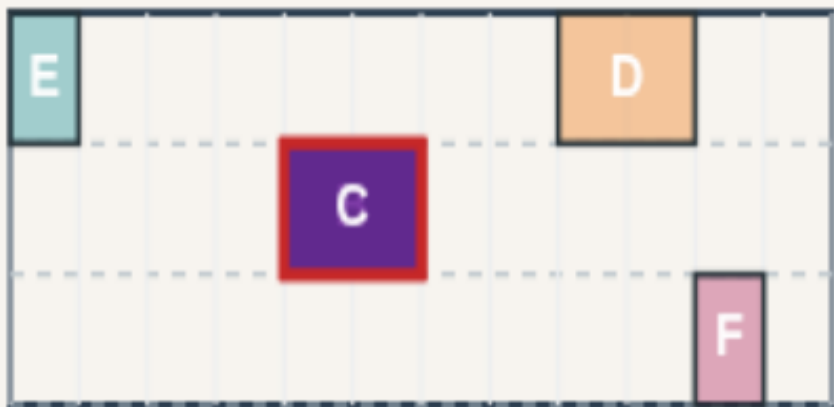
Start from overlap seed

ABACUS ROW PACK

Start from overlap seed

Step 1 / 5 · overlap-seed · module02-05-abacus-row-pack

chip 12x6 · 3 rows · site 1



Explanation

Abacus-lite processes movables by increasing original x. From the same overlap seed as overlap removal, it tries each row and picks the placement with minimum L1 displacement.

- Order: A, B, C, ... by x
- Try every row for each cell
- Leftmost legal site on row

```
seed: OVERLAP_PLACEMENT  
legal: false
```

Start from overlap seed

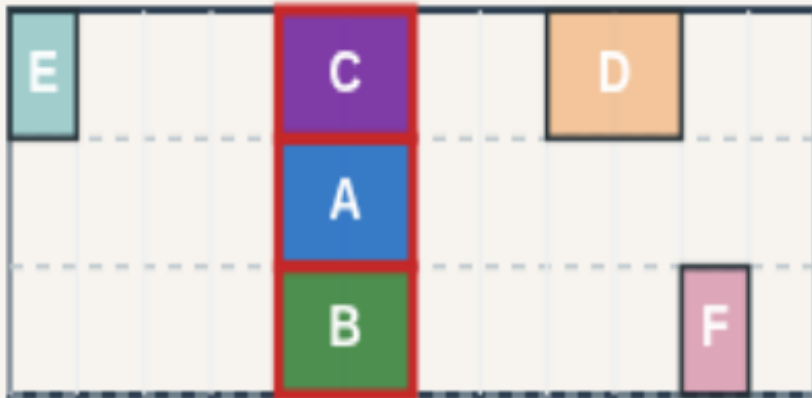
Process by x: try each row

ABACUS ROW PACK

Process by x: try each row

Step 2 / 5 · try-rows · module02-05-abacus-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Cell A tries rows zero, two, four—picks row two at x four (zero displacement). B chooses row zero at x four. C chooses row four at x four. Cross-row spread beats single-row pack.

- Per-cell row trial
- Min L1 from origin
- Tie-break: smaller x

A → row 2
B → row 0
C → row 4

Process by x: try each row

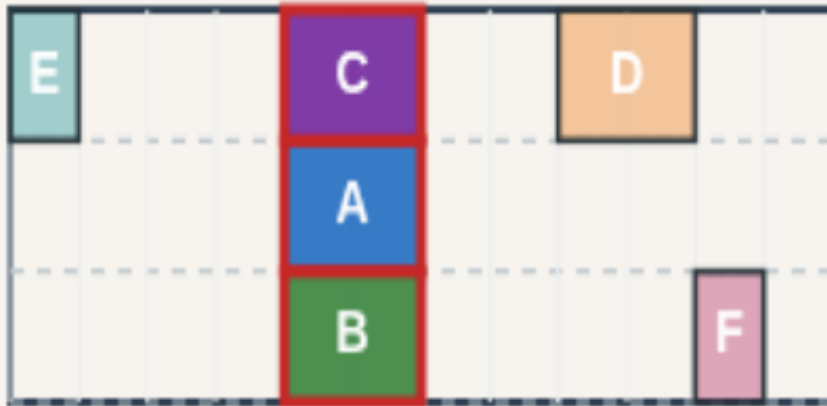
Result: A@4,2 B@4,0 C@4,4

ABACUS ROW PACK

Result: A@4,2 B@4,0 C@4,4

Step 3 / 5 · abacus-result · module02-05-abacus-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Final Abacus layout: A on middle row, B on bottom, C on top—all at x equals four but on different rows. D, E, F stay at their seed coordinates.

- A @ (4,2)
- B @ (4,0)
- C @ (4,4)

```
disp: 4
HPWL: 38
legal: true
```

Result: A@4,2 B@4,0 C@4,4

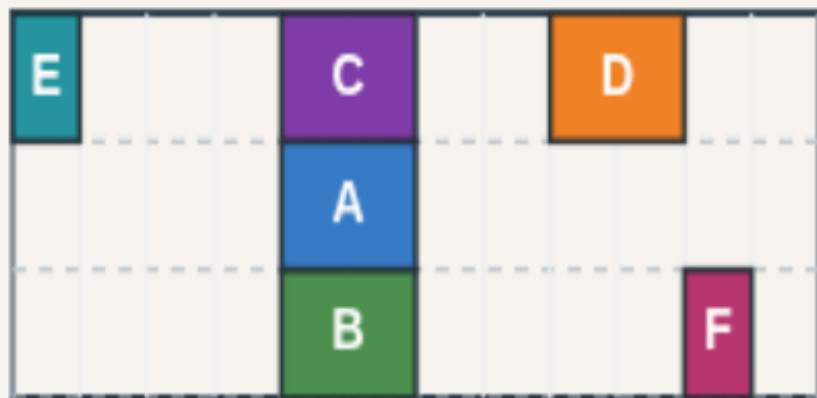
Lower displacement than Tetris

ABACUS ROW PACK

Lower displacement than Tetris

Step 4 / 5 · vs-overlap-removal · module02-05-abacus-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Abacus displacement is four versus six for overlap removal / Tetris on this seed. You pay three HPWL points—thirty-eight versus thirty-two—for staying closer to global targets.

- abacus disp: 4
- tetris disp: 6
- HPWL tradeoff: 38 vs 32

```
abacusDisp: 4  
tetrisDisp: 6  
abacusHpwl: 38
```

Lower displacement than Tetris

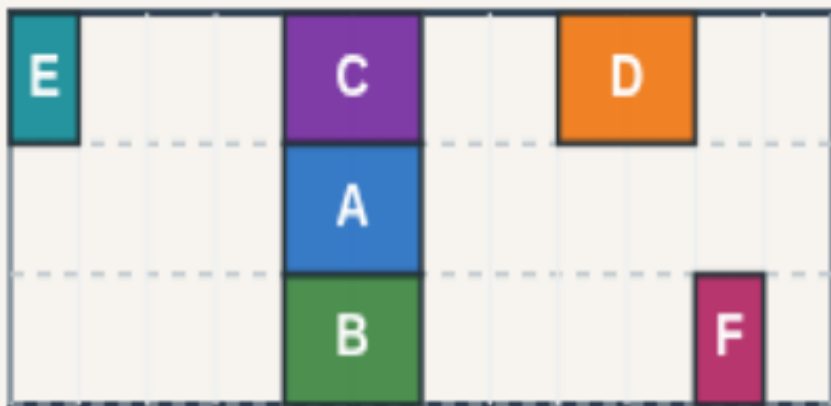
Abacus minimizes movement

ABACUS ROW PACK

Abacus minimizes movement

Step 5 / 5 · takeaway · module02-05-abacus-row-pack

chip 12×6 · 3 rows · site 1



Explanation

Abacus is the detailed legalizer in this course: try rows, pick min displacement. Use it when staying near global coordinates matters more than the last HPWL point.

- Cross-row assignment
- disp 4 golden
- Next: Tetris contrast

abacusLegal: true

Abacus minimizes movement

Browser lab track

- In the browser lab track, open the **abacus-row-pack** lab from the tools shelf
- Open the interactive lab
- Reveal golden is study-only
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny\_legal.json`, run the algorithm with deterministic coordinates
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module